

COURSE: CHE 312: Transport Phenomena II

Spring Semester 2020

Department of Chemical Engineering

University of Illinois at Chicago

Midsemester Exam #1

Only one sheet of paper 5 x 7 is allowed

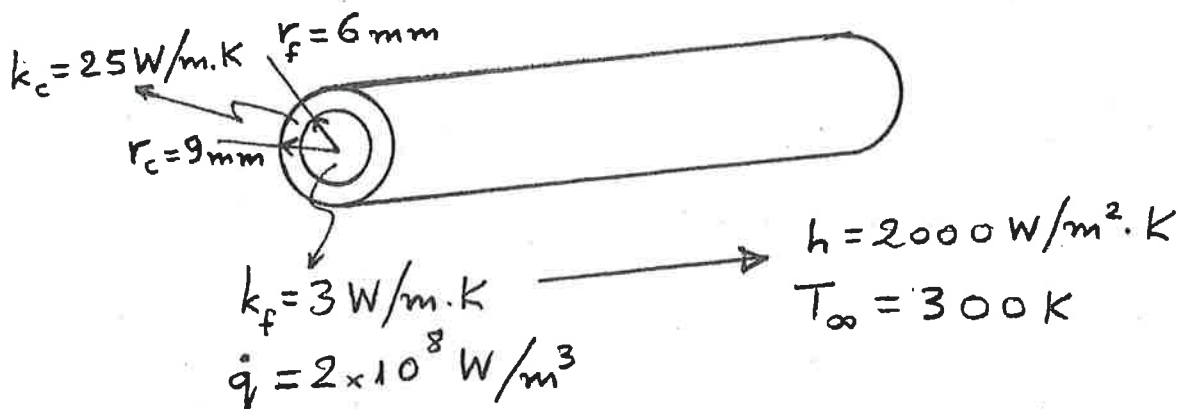
Duration: 50 minutes --- February 10, 2020

Both Problems must be done

Problem 1: (50 points)

A nuclear reactor fuel element consists of a solid cylindrical pin of radius r_f and thermal conductivity k_f . The fuel pin is in good contact with a cladding material of outer radius r_c and thermal conductivity k_c . Consider steady-state conditions for which uniform heat generation occurs within the fuel at a volumetric rate $\dot{q}$ and the outer surface of the cladding is exposed to a coolant that is characterized by a temperature T_∞ and a convection coefficient h .

- (10/50)
- Obtain equations for the temperature distributions $T_f(r)$ and $T_c(r)$ in the fuel and cladding, respectively. Express your results exclusively in terms of the foregoing variables.
 - Consider a uranium oxide fuel pin for which $k_f = 3 \text{ W/m}\cdot\text{K}$ and $r_f = 6 \text{ mm}$ and cladding for which $k_c = 25 \text{ W/m}\cdot\text{K}$ and $r_c = 9 \text{ mm}$. If $\dot{q} = 2 \times 10^8 \text{ W/m}^3$, $h = 2000 \text{ W/m}^2\cdot\text{K}$, and $T_\infty = 300 \text{ K}$, what is the maximum temperature in the fuel element?



Fuel $\frac{1}{r} \frac{\partial}{\partial r} (kr \frac{\partial T}{\partial r}) + \dot{q} = 0$

$\frac{\partial}{\partial r} (kr \frac{\partial T}{\partial r}) + \dot{q}r = 0$

$kr \frac{\partial T}{\partial r} = -\frac{\dot{q}r^2}{2} + C_1$

$\frac{\partial T}{\partial r} = -\frac{\dot{q}r}{2} + \frac{C_1}{r}$

$T(r) = -\frac{\dot{q}r^2}{4} + C_1 \ln(r) + C_2$ ✓

@ $T_s = -\frac{\dot{q}r^2}{4} + C_1 \ln(0) + C_2$

$C_2 = T_s$

$-k \left[\frac{\dot{q}r}{2} + \frac{C_1}{r} \right]_{r=r_f} = \dot{q}_s$

$-\frac{k\dot{q}r}{2} - \frac{kC_1}{r} = \dot{q}_s$

$-\frac{kC_1}{r} = \dot{q}_s + \frac{k\dot{q}r}{2}$

$-kC_1 = r_f \left[\dot{q}_s + \frac{k\dot{q}r_f}{2} \right]$

$C_1 = -\frac{r_f \left[\dot{q}_s + \frac{k\dot{q}r_f}{2} \right]}{k}$

Fuel $T(r) = -\frac{\dot{q}r^2}{4} - \frac{r_f \left[\dot{q}_s + \frac{k\dot{q}r_f}{2} \right]}{k} \ln(r) + T_s$

Cladding $T(r) = -\frac{\dot{q}r^2}{4} + C_1 \ln(r) + C_2$

$\frac{\partial T}{\partial r} = -\frac{\dot{q}r}{2} + \frac{C_1}{r}$

$-k \frac{\partial T}{\partial r} = \frac{k\dot{q}r}{2} + \frac{kC_1}{r}$

$C_2 = \dot{q}_s$

$C_1 = +\frac{\dot{q}r^2}{4} - \ln(r)$

Boundary Condition:

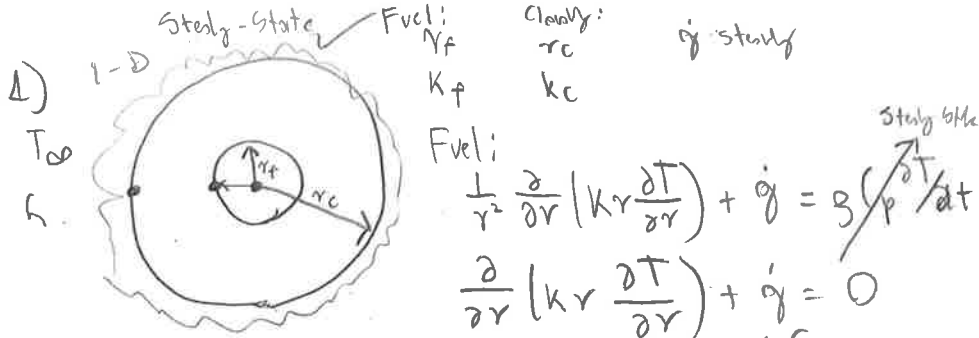
① $r=0$
② $r=r_f$

✓ $T(0, r) = T_s$
✗ $-k \frac{\partial T}{\partial r} \big|_{r=r_f} = \dot{q}_s$

Boundary Condition

① $r=r_f$ $-k \frac{\partial T}{\partial r} \big|_{r=r_f} = \dot{q}_s$
② $r=r_c$ $-k \frac{\partial T}{\partial r} \big|_{r=r_c} = h(T_\infty - T_c)$

✗



$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(k r \frac{\partial T}{\partial r} \right) + \dot{q} = 0$$

$$\frac{\partial}{\partial r} \left(k r \frac{\partial T}{\partial r} \right) + \dot{q} = 0$$

bc: @ $r=r_f$ $+k \frac{\partial T}{\partial r} \big|_{r=r_f} = \dot{q}_s$ $k r \frac{\partial T}{\partial r} = -\dot{q} r$

@ $r=0$ $T(0, t) = T_s$

$$\frac{\partial T}{\partial r} = -\frac{\dot{q} r}{k r}$$

$$\frac{\dot{q} r}{k} \ln(r) = T_s + \dot{q} + C_2$$

$$T(r) = -\frac{\dot{q} r}{k} \ln(r) + C_2$$

$$T_s + \dot{q} - C_2$$

$$C_1 = -\frac{T_s - C_2}{k \ln(r)} = \frac{C_2 - T_s}{k \ln(r)} = \frac{1}{e^{\ln(r)}} \left(\frac{C_2 - T_s}{k} \right)$$

no

$$C_1 = r e^{\frac{C_2 - T_s}{k}}$$

$$C_2 = \frac{\dot{q} r}{r}$$

Fuel

$$\frac{\partial}{\partial r} \left(k r \frac{\partial T}{\partial r} \right) = -\dot{q}$$

$$k r \frac{\partial T}{\partial r} = -\dot{q} r + C_1$$

$$\frac{\partial T}{\partial r} = -\dot{q} + \frac{C_1}{r}$$

$$T(r) = -\dot{q} r + \ln(r) C_1 + C_2$$

$$T_s = -\dot{q} r + \ln(r) C_1 + C_2$$

$$C_2 = T_s$$

$$C_1 = \frac{T_s - (-\dot{q} r + \ln(r) C_1 + C_2)}{\ln(r)}$$

Boundary Condition

@ $r=r_f$ $-k \frac{\partial T}{\partial r} \big|_{r=r_f} = \dot{q}_s$

@ $r=r_c$ $-k \frac{\partial T}{\partial r} \big|_{r=r_c} = h(T_\infty - T_{s,c})$

$$-k \left(-\dot{q} + \frac{C_1}{r} \right) = \dot{q}_s$$

$$k \dot{q} + \frac{k C_1}{r} = \dot{q}_s$$

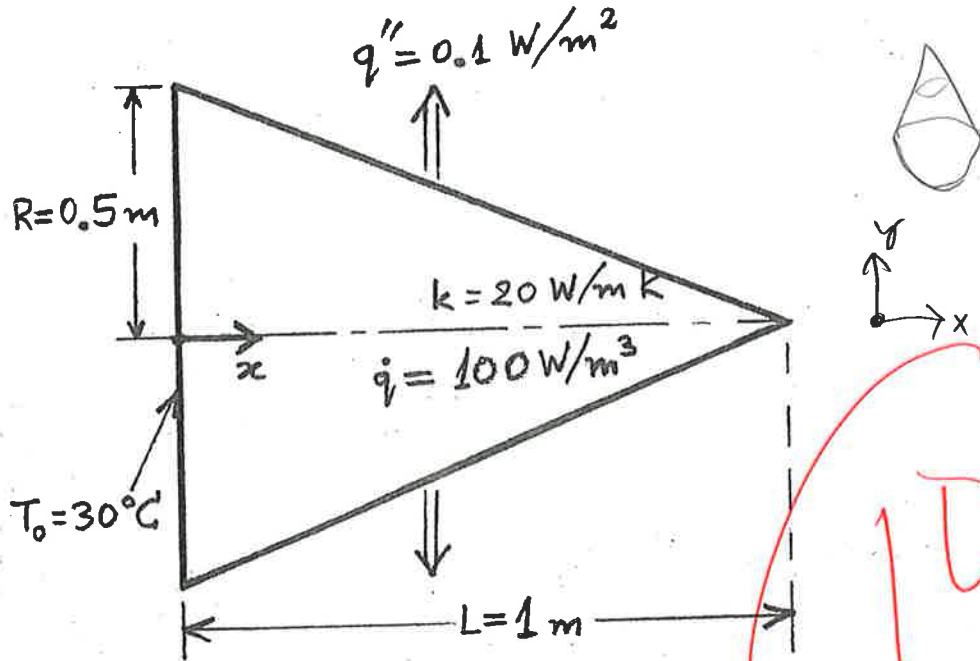
$$-\frac{k C_1}{r} = \dot{q}_s - k \dot{q}$$

$$-k C_1 = \dot{q}_s r - k \dot{q} r$$

$$C_1 = -\left(\frac{\dot{q}_s r - k \dot{q} r}{k} \right)$$

Problem 2: (50 points)

A solid cone as shown is of circular cross section and the radius of its base is 0.5 m. The cone of length 1 m and thermal conductivity 20 W/m·K has uniform volumetric heat generation of 100 W/m³. A constant heat flux of 0.1 W/m² is leaving the cone through the lateral surface of the cone, while the end surface at $x = 0$ is maintained at $T_0 = 30^\circ\text{C}$. Determine the temperature profile $T(x)$ and the temperature $T(L)$.



$$\begin{aligned} R &= 0.5 \text{ m} \\ L &= 1 \text{ m} \\ k &= 20 \text{ W/m}\cdot\text{K} \\ \dot{q} &= 100 \text{ W/m}^3 \\ \dot{q}'' &= 0.1 \text{ W/m}^2 \end{aligned}$$

$$T(0, y) = 30^\circ\text{C}$$

$$\Theta = \frac{T - T_1}{T_2 - T_1}$$

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\dot{q}}{k} = 0 \Rightarrow \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = -\frac{\dot{q}}{k}$$

$$\Theta(x, y) = X(x) \cdot Y(y) \Rightarrow -\frac{1}{X} \frac{d^2 X}{dx^2} = \frac{1}{Y} \frac{d^2 Y}{dy^2}$$

$$\Theta = [C_1 \cos \lambda x + C_2 \sin \lambda x] \cdot [C_3 e^{-\lambda y} + C_4 e^{\lambda y}]$$

$$\Theta(x, y) = \frac{2}{n} \sum \frac{(-1)^{n+1} + 1}{n} \sinh \frac{n\pi x}{L} \frac{\sinh(n\pi y/L)}{\sinh(n\pi W/L)}$$

